The Galerkin method

subspaces of V with the following properties:

a) every V_k is *finite-dimensional*: $\dim V_k = k$;

b) $V_k \subset V_{k+1}$ (actually, not strictly necessary);

c) $\overline{\cup V_k} = V$.

To realize the projection, assume that the vectors $\psi_1, \psi_2, \ldots, \psi_k$ span V_k. Then, we look for an approximation of the solution u in the form

$$u_k = \sum_{j=1}^{k} c_j \psi_j, \tag{7.33}$$

by solving the *projected* problem

$$a\left(u_k, v\right) = \langle F, v \rangle_* \qquad \forall v \in V_k. \tag{7.34}$$

The Galerkin Method

$$\text{find } u_k \in V_k : a(u_k, v_k) = F(v_k) \quad \forall v_k \in V_k$$

Is it a well posed problem?

The Galerkin Method

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Is it a well posed problem?

Lemma 3.1 (di Lax-Milgram) *Sia V uno spazio di Hilbert, $a(\cdot,\cdot) : V \times V \to \mathbb{R}$ una forma bilineare continua e coerciva, $F(\cdot) : V \to \mathbb{R}$ un funzionale lineare e limitato. Allora esiste unica la soluzione del problema:*

$$\text{trovare } u \in V : a(u, v) = F(v) \ \forall v \in V. \tag{3.38}$$

The Galerkin Method

$$\text{find } u_k \in V_k : a(u_k, v_k) = F(v_k) \quad \forall v_k \in V_k$$

Is it a well posed problem?

Theorem 7.10 (Lax-Milgram). *Let V be a real Hilbert space endowed with inner product $(\cdot,\cdot)$ and norm $\|\cdot\|$. Let $a = a(u,v)$ be a bilinear form in V. If:*

i) *a is **continuous**, i.e. there exists a constant M such that*

$$|a(u,v)| \leq M \|u\| \|v\|, \qquad \forall u, v \in V;$$

ii) *a is $V-$**coercive**, i.e. there exists a constant $\alpha > 0$ such that*

$$a(v,v) \geq \alpha \|v\|^2, \qquad \forall v \in V, \tag{7.24}$$

then there exists a unique solution $\overline{u} \in V$ of problem (7.23). Moreover, the following stability estimate holds:

$$\|\overline{u}\| \leq \frac{1}{\alpha} \|F\|_{V^*}. \tag{7.25}$$

The Galerkin Method

$$\mathrm{A}_{\mathrm{G}}\mathbf{u} = \mathbf{f}_{\mathrm{G}}.$$

is it a solvable linear system?

does it have a unique solution?

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Teorema 4.1 *La matrice* A *associata alla discretizzazione del problema ellittico (4.1) con forma bilineare* $a(\cdot,\cdot)$ *coerciva con il metodo di Galerkin è definita positiva.*

Proprietà 4.1 *La matrice* A *è simmetrica se e solo se la forma bilineare* $a(\cdot,\cdot)$ *è simmetrica.*

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Corollario 4.1 *La soluzione del problema di Galerkin* (4.2) *esiste ed è unica.*

The Galerkin Method

Matrix B is SPD if: $\mathbf{v}^T \mathrm{B} \mathbf{v} \geq 0 \; \forall \mathbf{v} \in \mathbb{R}^N \text{ and that } \mathbf{v}^T \mathrm{B} \mathbf{v} = 0 \Leftrightarrow \mathbf{v} = \mathbf{0}$

A_G is SPD - Proof:

The Galerkin Method

Matrix B is SPD if: $\mathbf{v}^T \mathrm{B} \mathbf{v} \geq 0 \; \forall \mathbf{v} \in \mathbb{R}^N$ and that $\mathbf{v}^T \mathrm{B} \mathbf{v} = 0 \Leftrightarrow \mathbf{v} = \mathbf{0}$

$\mathbf{A_G}$ is SPD - Proof:

Let us associate with a generic vector $\mathbf{v} = (v_i)$ of $\mathbb{R}^N$ the function $v_h = \sum_{j=1}^N v_j \varphi_j \in V_h$. Since the form $a(\cdot, \cdot)$ is bilinear and coercive we get

$$\begin{aligned}
\mathbf{v}^T \mathrm{A_G} \mathbf{v} &= \sum_{j=1}^N \sum_{i=1}^N v_i a_{ij} v_j = \sum_{j=1}^N \sum_{i=1}^N v_i a(\varphi_j, \varphi_i) v_j \\
&= \sum_{j=1}^N \sum_{i=1}^N a(v_j \varphi_j, v_i \varphi_i) = a\left(\sum_{j=1}^N v_j \varphi_j, \sum_{i=1}^N v_i \varphi_i \right) \\
&= a(v_h, v_h) \geq \alpha \|v_h\|_V^2 \geq 0.
\end{aligned}$$

Moreover, if $\mathbf{v}^T \mathrm{A_G} \mathbf{v} = 0$ then also $\|v_h\|_V^2 = 0$ which implies $v_h = 0$ and thus $\mathbf{v} = \mathbf{0}$.

Analysis of the Galerkin method

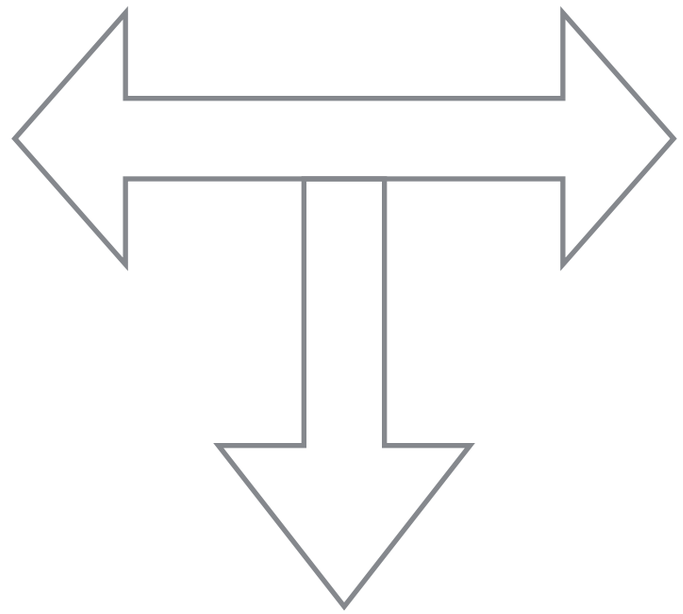
$$\text{find } u_h \in V_h : \; a(u_h, v_h) = (f, v_h) \quad \forall v_h \in V_h.$$

(consistency)

(stability)

(convergence)

(consistency)



(stability)

(convergence)

Consistency

$$\text{find } u \in V: \; a(u,v) = (f,v) \;\; \text{for all } v \in V,$$

$$\boldsymbol{u}$$

$$\text{find } u_h \in V_h: \; a(u_h,v_h) = (f,v_h) \quad \forall v_h \in V_h.$$

$$a(u,v_h) - (f,v_h) = ??$$

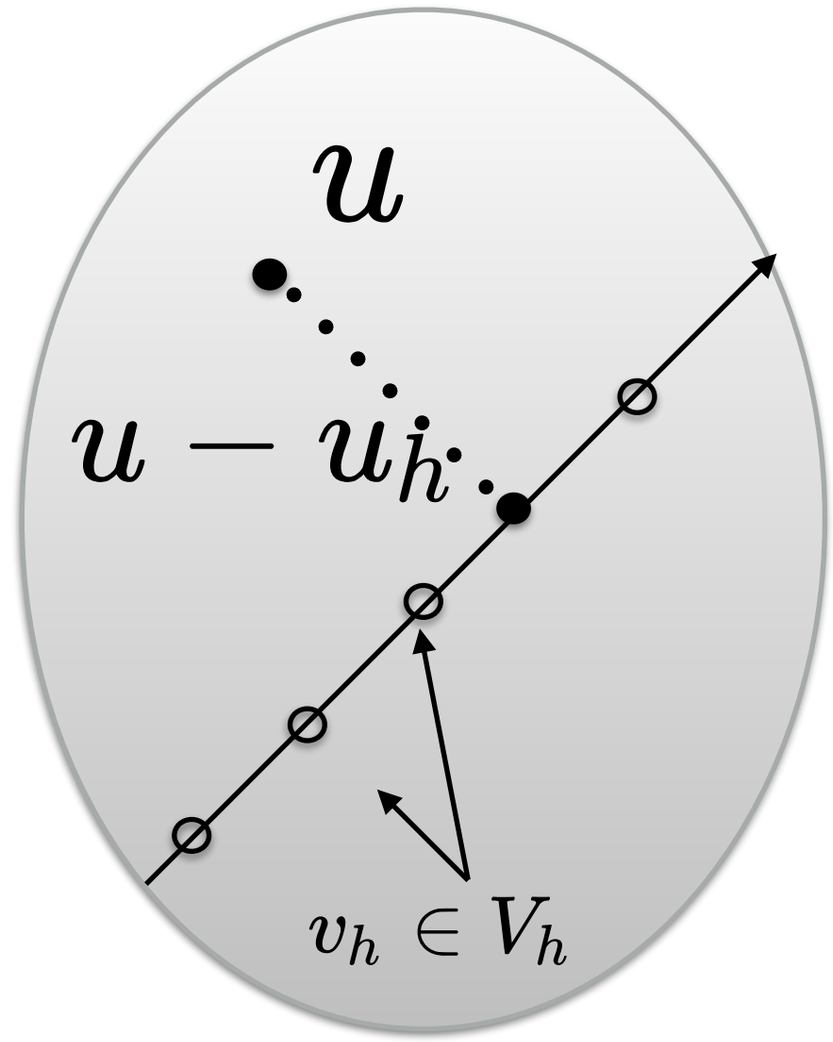
$$V_h \subset V$$

Galerkin Orthogonality

$$a(u, v_h) - (f, v_h) = 0 \; \forall \; v_h \in V_h$$

$$a(u, v_h) - a(u_h, v_h) = 0 \; \forall \; v_h \in V_h$$

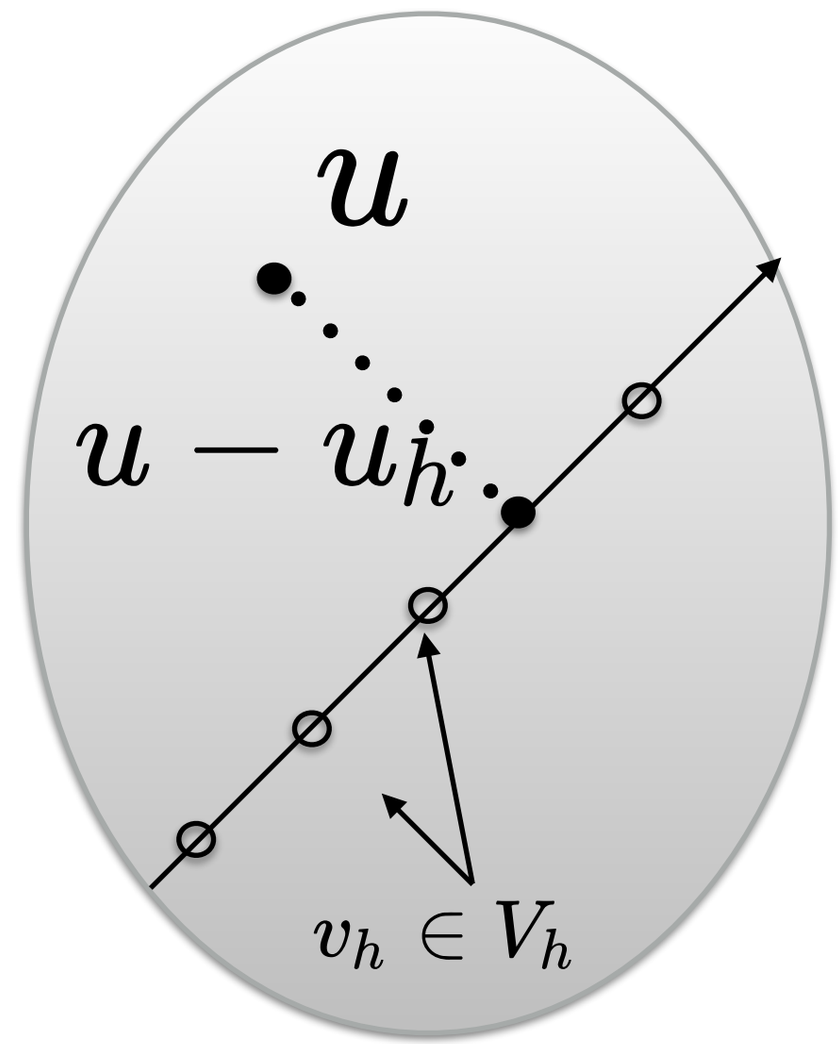
$$a(u - u_h, v_h) = 0 \; \forall \; v_h \in V_h$$



Galerkin Orthogonality

$$a(u - u_h, v_h) = 0 \; \forall \; v_h \in V_h$$

The Galerkin solution u_h is the (Galerkin) projection of u on V_h



If a(.,.) is a scalar product then

$$a(v_h, v_h) = \|v_h\|_a^2$$

is a norm, the energy norm ….

Galerkin Orthogonality

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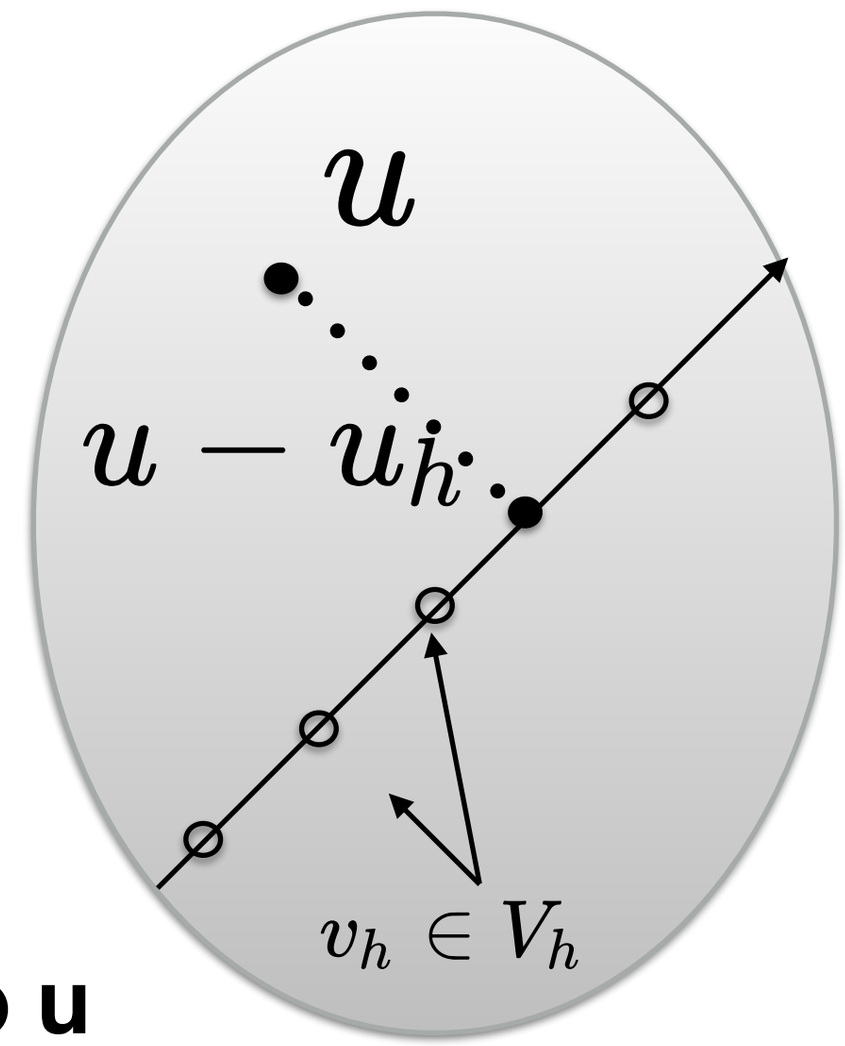


The Galerkin solution u_h is the (Galerkin) projection of u on V_h

If a(.,.) is a scalar product then

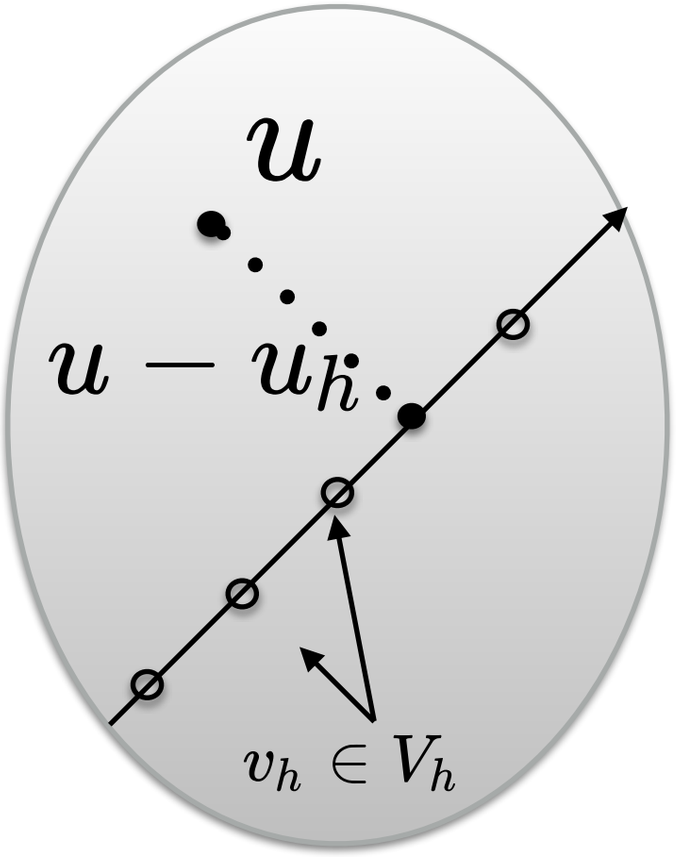
$$a(v_h, v_h) = \|v_h\|_a^2$$

is a norm, the energy norm,

then u_h is the closest element of V_h to u

$$\|u - u_h\|_a = \inf_{v_h \in V_h} \|u - v_h\|_a$$

Galerkin Orthogonality



Lemma 4.1 (ortogonalità di Galerkin) *La soluzione u_h del metodo di Galerkin soddisfa la proprietà*

$$a(u - u_h, v_h) = 0 \; \forall v_h \in V_h. \tag{4.8}$$

Stability

$$a(u_h, v_h) = (f, v_h)\ \forall v_h \in V_h$$

$$a(u_h, u_h) = (f, u_h)$$

$$\alpha \|u_h\|_1^2 \leq a(u_h, u_h) \leq \|f\|_0 \|u_h\|_0$$

$$\alpha \|u_h\|_1^2 \leq a(u_h, u_h) \leq \|f\|_0 \|u_h\|_1$$

$$\|u_h\|_1 \leq \frac{1}{\alpha} \|f\|_0$$

The Galerkin method is stable, uniformly with respect to h

Stability

Corollario 4.2 *Il metodo di Galerkin è stabile, uniformemente rispetto ad h, in quanto vale la seguente maggiorazione della soluzione*

$$\|u_h\|_V \leq \frac{1}{\alpha}\|F\|_{V'}.$$

(consistency) (stability)

(convergence)

Lemma di Cea'

$$\|u - u_h\|_1 \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_1$$

CEA, J. Approximation variationnelle des problemes aux limites. *Ann*. Inst. Fourier, Grenoble **14** (1964) 345-444.

Convergence

$$a(u-u_h,u-u_h)=a(u-u_h,u\pm v_h-u_h)=a(u-u_h,u-v_h)+a(u-u_h,v_h-v_h)$$

$$\alpha\|u-u_h\|_1^2\leq a(u-u_h,u-u_h)=a(u-u_h,u-v_h)\leq M\|u-u_h\|_1\|u-v_h\|_1$$

$$\|u-u_h\|_1\leq\frac{M}{\alpha}\|u-v_h\|_1\ \forall v_h\in V_h$$

Convergence

$$\|u - u_h\|_1 \leq \frac{M}{\alpha} \|u - v_h\|_1 \; \forall v_h \in V_h$$

Lemma di Cea

$$\|u - u_h\|_1 \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_1$$

Compare with Galerkin projection

$$\|u - u_h\|_a = \inf_{v_h \in V_h} \|u - v_h\|_a$$

Convergence

Lemma di Cea

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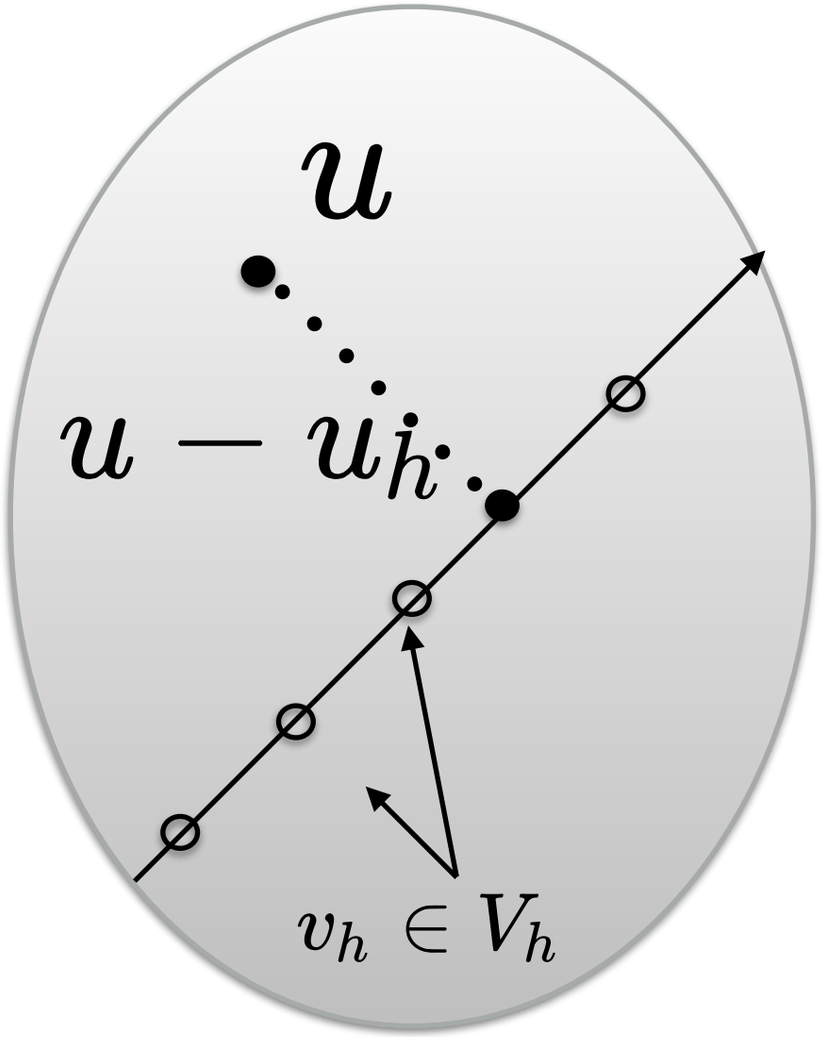

Compare with Galerkin projection

$$\|u - u_h\|_a = \inf_{v_h \in V_h} \|u - v_h\|_a$$